

LIFTING AND HANDLING MANUAL

VOLUME I – CHAPTER 8

6.1 CONTRACTOR CRANE, MULTI-PURPOSE MACHINE, FORKLIFT, CONSTRUCTION EQUIPMENT, AND RIGGING GEAR REQUIREMENTS

6.1.1 The following is a list of requirements that contractors shall comply with for all contracts that may result in the use of a Category 1 or 4 crane, multi-purpose machines, forklifts, construction equipment, and rigging gear when used on Navy property to lift suspended loads. Non-compliance with the requirements of this instruction may result in denial of access, stopping of operations, or removal from shipyard property.

6.1.2 References

6.1.2.1 NAVFAC P-307, Weight Handling Program Management

6.1.2.2 American Society of Mechanical Engineers (ASME) B30.3 (tower cranes), B30.5 (mobile cranes), B30.8 (floating cranes), B30.9 (slings), B30.20 (below the hook lifting devices), B30.22 (articulating booms), B30.26 (rigging hardware); ANSI/ITSDF B56.6 (rough terrain forklifts); Safety Standards for Cableways, Cranes, Derricks, Hoists, Hooks, Jacks, and Slings

6.1.2.3 CFR, Title 29, Chapter XVII, Part 1917, Marine Terminals.

6.1.2.4 CFR, Title 29, Chapter XVII, Part 1926, Safety and Health Regulations for Construction

6.1.2.5 CFR, Title 29, Chapter XVII, Part 1915, Occupational Safety and Health Standards for Shipyard Employment

6.1.2.6 OPNAVINST 5100.23, Navy Safety and Occupational Health Program Manual

6.1.2.7 EM 385-1-1, Safety and Health Requirements Manual, U.S. Army Corps of Engineers

6.1.2.8 NAVFAC Guide Specification NFGS-01525D, Safety Requirements

6.1.3 These requirements are solely intended to provide for the protection of Government property and personnel and are not intended to, and do not, in any manner whatsoever, relieve the contractor of its responsibility, including, without limitation, its responsibility for the protection of its equipment and personnel.

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6.1.4 Notification Requirement. Contractor shall notify the contracting officer in advance of the intent of bringing a non-Navy owned crane onto Shipyard property (allows contracting officer to notify shipyard 3 days in advance of entry), or of any multi-purpose machines, material handling equipment, or construction equipment that may be used in a crane-like application to lift suspended loads. The contractor shall also specify when crane entry onto Shipyard property is scheduled during back shift, weekend, or holiday hours of operation. All entries shall be through a prearranged entry point. The following documentation shall be provided along with notification: a copy of the Certification of Compliance (reference 6.1.2.1) and objective evidence of operator qualifications for cranes with rated capacities of 2,000 lbs. or greater. Failure to schedule or provide necessary documentation may result in the crane being denied access to the facility.

6.1.5 The contractor shall comply with applicable reference 6.1.2.2 standards (e.g., B30.3 for construction tower cranes, B30.5 for mobile cranes, B30.8 for floating cranes, B30.9 for slings, B30.20 for below the hook lifting devices, and B30.22 for articulating boom cranes), B30.26 for rigging hardware, and ANSI/ITSDF B56.6 for rough terrain forklifts). Floating cranes and barge-mounted mobile cranes require the following:

- Third party certification from an OSHA accredited organization (or from a state accredited organization for those states with OSHA approved state plans). Third party certification is not required for floating cranes and barge-mounted mobile cranes at Navy activities in foreign countries.
- A load indicating device, a wind-speed indicating device, and a marine type list and trim indicator readable in one-half degree increments.

6.1.6 Certification of Compliance (reference 6.1.2.1). The contractor shall complete a certificate of compliance that the crane (or other machine if used to lift suspended loads) and rigging gear meet applicable OSHA and ANSI/ASME regulations (with the contractor citing which OSHA regulations are applicable, e.g., cranes/multi-purpose machines used in cargo transfer shall comply with reference 6.1.2.3; cranes/multi-purpose machines used in construction, demolition, or maintenance shall comply with reference 6.1.2.4; cranes/multi-purpose machines used in ship repair or shipbreaking shall comply with reference 6.1.2.5; aerial lifts/platforms (sign trucks) with hoists shall comply with 29 CFR 1926.453 and ANSI A92.2; personnel hoists shall comply with 29 CFR 1926.552 and ASSE A10.22; slings shall comply with ASME B30.9; rigging hardware shall comply with ASME B30.26). For cranes (or other machines used to lift suspended loads) and rigging equipment at naval activities in foreign countries, the contractor shall certify that the crane (or other machine) and the rigging equipment conform to the appropriate host nation safety standards. The contractor shall also certify that all of its crane (or other machine) safety devices are enabled and functioning properly, that all personnel working on the naval activity job site have been trained to not stand under or in the fall zone of a suspended load unless specifically allowed by reference 6.1.2.7, and that

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operators have been trained to not bypass safety devices (e.g., anti-two block devices, LMIs) during lifting operations. Certifications are required to be available on the crane or in the contractor's on-site office for rigging operations. When a crane on shipyard property is not authorized for use, the Certification of Compliance shall state, "Operation of this Crane is NOT Authorized."

6.1.7 The contractor shall certify (reference 6.1.2.1) that the crane or machine operator is qualified and trained for the operation of the crane to be used. For mobile cranes, commercial truck mounted cranes, and articulating boom cranes with OEM rated capacities of greater than 2,000 pounds, require that the crane operator be designated as qualified by a source that qualifies crane operators (i.e., a union, a government agency, or an organization that tests and qualifies crane operators). Operators of cranes used in construction at activities under US jurisdiction shall follow the qualification and certification requirements of 29 CFR 1926.1427. The contractor shall certify that signal persons used in construction work are qualified in accordance with 29 CFR 1926.1428. For Navy activities in foreign countries, follow host nation requirements as applicable in lieu of the CFRs. Proof of current qualification shall be provided.

6.1.8 For multi-purpose machines, material handling equipment, and construction equipment used to lift loads suspended by rigging equipment, the contractor shall have proof of authorization from the machine OEM that the machine is capable of making lifts of loads suspended by rigging equipment. Written approval from a qualified registered professional engineer, after a safety analysis is performed, is allowed in lieu of the OEM's approval. The contractor shall demonstrate that the equipment is properly configured to make such lifts and is equipped with a load chart. MHE attachments covered by and used in accordance with NAVSEA SW023-AH-WHM-010 have been pre-approved for use with all MHE. These attachments are exempt from the OEM authorization requirement.

6.1.9 All hooks used on cranes, hoists, other machines, and rigging gear shall have self-closing latches or the throat opening shall be "moused" (secured with wire, rope, heavy tape, etc.) or otherwise secured to prevent the attached item from coming free of the hook under a slack condition. The following exceptions apply and shall be approved by the contractor's technical organization: items where the hook throat is fully obstructed and not available for manual securing and lifts where securing the hook throat increases the danger to personnel such as forge shop, dip tank, or underwater work.

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6.1.10 Loading Limitations

CAUTION: Piers and waterfront areas such as along dry docks and quay walls may have load restrictions.

6.1.10.1 The contractor shall notify the contracting officer prior to moving a crane on a pier, dry dock, or other waterfront area. The contractor shall provide the contracting officer with the crane make, model, and configuration in which it is to be used.

6.1.10.2 The contractor shall comply with crane access routes and load limitations issued with the contract.

6.1.11 Prior to making any critical lift, the contractor shall provide a critical lift plan for each of the following lifts: lifts over 75 percent of the capacity of the crane, hoist, or other machine (lifts over 50 percent of the capacity of a barge mounted mobile crane's hoists) at any radius of lift; lifts involving more than one crane, hoist, or other machine; lifts of personnel (lifts of personnel suspended by rigging equipment from multi-purpose machines, material handling equipment, or construction equipment shall not be permitted); lifts made in the vicinity of overhead power lines; erection of cranes; lifts of submerged or partially submerged objects; and lifts involving non-routine rigging or operation, sensitive equipment, or unusual safety risks. The plan shall include the following as applicable:

6.1.11.1 The size and weight of the load to be lifted, including crane (or other machine) equipment and rigging equipment that add to the weight. The OEM's maximum load capacities for the entire range of the lift shall also be provided.

6.1.11.2 The lift geometry, including the crane (or other machine) position, boom length and angle, height of lift, and radius for the entire range of the lift. Applies to both single and multiple-crane/machine lifts.

6.1.11.3 A rigging plan, showing the lift points, rigging equipment, and rigging procedures.

6.1.11.4 The environmental conditions under which lift operations are to be stopped.

6.1.11.5 For lifts of personnel, the plan shall demonstrate compliance with the requirements of 29 CFR 1926.1431. For Navy activities in foreign countries follow host nation requirements as applicable in lieu of the CFR.

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6.1.11.6 For barge mounted mobile cranes, barge stability calculations identifying crane placement/footprint; barge list and trim based on anticipated loading; and load charts based on calculated list and trim specific to the barge the crane is mounted on. The amount of list and trim shall be within the crane manufacturer's requirements.

6.1.11.7 For lifts in the vicinity of overhead power lines (i.e., if any part of the crane or other machine, including the fully extended boom of a telescoping boom crane or machine, or the load could approach the distances noted in figure 10-3 of reference 6.1.2.1 during a proposed operation), the plan shall demonstrate compliance to 29 CFR 1926.1408 - 1411. For Navy activities in foreign countries follow host nation requirements as applicable in lieu of the CFR.

6.1.12 Contractor provided tower cranes used in construction must meet the additional requirements of CFR 1926.1435. The following additional documentation is required for contractor provided tower cranes (those cranes defined by ASME B30.3).

6.1.12.1 Foundation design and requirements.

6.1.12.2 Installation instructions.

6.1.12.3 Assembly and disassembly instructions including climbing/jumping instructions if applicable.

6.1.12.4 Operating manual, limitations, and precautions.

6.1.12.5 Periodic inspection and maintenance requirements.

6.1.12.6 For tower cranes at Navy activities in foreign countries, the tower cranes shall comply with the appropriate host nation safety standards and industry consensus safety standards (e.g., EN or ISO standards).

6.1.13 Crane and Rigging Gear Accident and Near Miss Reporting and Record Keeping. Contractors operating cranes on Shipyard property shall report all WHE accidents and near misses that occur incidental to an operation, project, or facility as prescribed by paragraphs (6.1.13.1) through (6.1.13.5) requirements below. Contractors shall report directly to their respective contracting officer. There are two general categories of accidents as defined below. Crane accidents are those that occur during operation of a crane. Rigging gear accidents are those that occur when gear is used by itself in weight handling operation i.e., without a crane.

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6.1.13.1 Crane Accident: For the purpose of this definition, it is assumed there is an "operating envelope" around any crane. The operating envelope consists of any of the following elements:

- The crane (exception: a crane being operated in transit as defined by reference 6.1.2.1)
- The operator
- The riggers, signal persons, and crane walker
- Other personnel involved in the operation (supervisor, mechanic, tag line handler, engineer, etc.)
- The rigging gear between the hook and the load
- The load
- The crane's supporting structure (ground, rail, etc.)
- The lift procedure

6.1.13.1.1 Definition. A crane accident occurs when any of the elements in the crane operating envelope fails to perform correctly during a crane operation (see reference 6.1.2.1), including operation during maintenance, or testing resulting in any of the following:

- Personnel injury or death (Minor injuries that are inherent in any industrial operation, including strains and repetitive motion related injuries, shall be reported by the normal personnel injury reporting process in lieu of these requirements.)
- Material or equipment damage
- Dropped load (includes any part of the load or rigging gear and any item lifted with the load or rigging gear)
- Derailment
- Two-blocking
- Overload (includes load tests when the nominal test load is exceeded)
- Collision (avoidable contact between the load, crane, and/or other objects). Avoidable contact is defined as contact that would have been prevented with proper lift planning and execution. Simply briefing that contact will occur is not sufficient.

[NOTE: *[Bullets]* 3, 4, 5, 6, and 7 are considered crane accidents even though no material damage or injury occurs.

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6.1.13.1.2 A component failure (e.g., motor burnout, gear tooth failure, bearing failure) shall be considered an accident only if damage to the load or another crane component occurs as a result of the failure. An accident involving a mobile crane (Cat 1 or 4) that is configured for transit is considered an unplanned occurrence.

6.1.13.2 Rigging Gear Accidents: For the purpose of this definition, it is assumed there is an “operating envelope” around any rigging or other reference 6.1.2.1 section 14 equipment operation, and inside the envelope are the following:

- Rigging gear and miscellaneous equipment identified in reference 6.1.2.1 section 14
- The user of the gear or equipment (including operators of multi-purpose machines, MHE, and construction equipment)
- Other personnel involved in the operation (supervisor, mechanic, tagline handler, engineer, etc.)
- The load
- The gear or equipment’s supporting structure (padeyes, ship’s structure, building structure, etc.)
- The load’s rigging path
- The rigging procedure

6.1.13.2.1 Definition. A rigging accident occurs when any of the elements in the operating envelope fails to perform correctly during a rigging operation (see reference 6.1.2.1) resulting in any of the following:

- Personnel injury or death (Minor injuries that are inherent in any industrial operation, including strains and repetitive motion related injuries, shall be reported by the normal personnel injury reporting process of the activity in lieu of these requirements.)
- Material or equipment damage that requires the damaged item to be repaired because it can no longer perform its intended function. This does not include superficial damage such as scratched paint, minor lagging damage, or normal wear on rigging gear.
- Dropped load (includes any part of the load or rigging gear and any item lifted with the load or rigging gear)
- Two-blocking of cranes and powered hoists identified in section 14 of reference 6.1.2.1
- Overload. (includes load tests when the test load tolerance is exceeded)

[NOTE: Bullets] 3, 4, and 5 are considered accidents even though no material damage or injury occurs.

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6.1.13.2.2 A component failure (e.g., motor burnout, gear tooth failure, bearing failure) shall be considered an accident only if damage to the load or another component occurs as a result of the failure.

6.1.13.3 Near Miss: A near miss is an unplanned event during a weight handling operation that did not result in a definable accident noted above but easily had the potential to do so. Only a break in the chain of events prevented an accident. Simply put, a near miss is an accident that almost took place. The difference between a near miss and an accident (serious or otherwise) is often a fraction of an inch or a split second of time. A near-miss report is used to learn from situations where an accident “almost” happened so that the real event can be averted. By focusing on and learning from minor events, it is possible to reduce the probability of an actual accident from occurring. Near-miss reporting provides an opportunity to identify risks and mitigate the potential for an accident. It is a means for correcting issues that did not result in loss or injury, and it increases awareness of those personnel involved in or performing work. It is not intended that a near-miss investigation and report be as thorough as that for a crane or rigging accident; however, the investigation and report should be commensurate with the significance of the event.

6.1.13.4 An “unplanned occurrence” describes an event that does not meet the definition of a crane or rigging accident but results in injury, or damage to a crane, crane component, or related equipment due to an event not directly related to a weight handling operation. Examples include, but are not limited to, injury or damage caused by weather, damage to a parked or stationary crane caused by another moving object (e.g. vehicle, forklift), significant maintenance errors, and flooding or fire damage.

6.1.13.5 The contractor shall notify the contracting officer as soon as practical, but not later than four hours, after any WHE accident. The contractor shall secure the accident site and protect evidence until released by the contracting officer. The contractor shall conduct an investigation to establish the root cause(s) of any WHE accident, near miss, or unplanned occurrence. Crane operations shall not proceed until the cause is determined and corrective actions have been implemented to the satisfaction of the contracting officer. The contractor shall certify (using figure P-1 of reference 6.1.2.1) that the operators, riggers and company officials are aware of the actions required in the event of an accident as specified in the contract. The contractor shall provide the contracting officer within 30 days of any accident or near miss a Crane and Rigging Accident Report or a Near Miss and Unplanned Occurrence Report using the form provided in reference 6.1.2.1 consisting of a summary of circumstances, an explanation of causes(s), photographs if available, and corrective actions taken. These notifications and reporting requirements are in addition to those promulgated by reference 6.1.2.6 and related claimant instructions.

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6.1.14 The contractor shall certify (using figure P-1 of reference 6.1.2.1) that the riggers are qualified to perform the work by either signing a statement of compliance, providing proof of successful completion of a rigging apprenticeship by a union or other organization providing apprenticeship training, or proof of successful completion of training courses from a recognized source such as a trade union, a government agency, or an organization that tests and qualifies riggers.

6.1.15 Refer to reference 6.1.2.1 for additional requirements which apply to contracts involving work on telecommunication towers or with personnel hoists on telecommunication towers.

6.1.16 Each contractor shall perform the following actions prior to conducting crane operations on Shipyard property:

6.1.16.1 Inspection Requirements. It shall be the sole responsibility of the contractor to assure the contracting officer and/or designated Shipyard personnel that the crane and associated rigging gear are in good working order and safe for use.

6.1.16.1.1 Crane Inspection. Perform pre-operational inspection of the crane in the presence of a representative of the contracting office of the crane prior to starting work on Shipyard property. Inspection shall meet all applicable reference 6.1.2.2, reference 6.1.2.7 (for NAVFAC construction contracts), and OSHA requirements.

6.1.16.1.2 Wire Rope Inspection. Perform a Wire Rope Inspection in the presence of a representative of the contracting office to applicable reference 6.1.2.2, reference 6.1.2.7 (for NAVFAC construction contracts), and OSHA requirements.

6.1.16.1.3 Rigging Gear Inspection. Perform a Rigging Gear Inspection in the presence of a representative of the contracting office to applicable reference 6.1.2.2, reference 6.1.2.7 (for NAVFAC construction contracts), and OSHA requirements.

**APPENDIX P CONTRACTOR CRANE (OR ALTERNATE MACHINE USED TO LIFT
SUSPENDED LOADS) AND RIGGING GEAR REQUIREMENTS**

CERTIFICATE OF COMPLIANCE	
This certificate shall be signed by an official of the company that provides cranes (or multi-purpose machines, MHE, or construction equipment used to lift loads suspended by rigging gear) or rigging gear for any application under this contract. Post a completed certificate on each crane or alternate machine (or in the contractor's on-site office for rigging operations) brought onto Navy property.	
CONTRACTING OFFICER'S POINT OF CONTACT (Government Representative)	PHONE
PRIME CONTRACTOR/PHONE	CONTRACT NUMBER
CRANE OR ALTERNATE MACHINE SUPPLIER/PHONE (if different from prime contractor)	CRANE OR ALTERNATE MACHINE NUMBER (i.e., ID number)
CRANE OR ALTERNATE MACHINE MANUFACTURER/TYPE/CAPACITY	
CRANE OR ALTERNATE MACHINE OPERATOR'S NAME(S)	
I certify that 1. The above noted crane or alternate machine and all rigging gear conform to applicable OSHA regulations (host nation regulations for naval activities in foreign countries) and applicable ASME B30 or other standards. The following OSHA regulations and ASME or other standards apply: 2. The operators noted above have been trained and are qualified, including physical qualifications, for the operation of the above noted crane(s) or alternate machine(s). 3. All operational safety devices and operator aids are enabled and functioning properly and the operators noted above have been trained not to bypass operational safety devices and operator aids during lifting operations. 4. The operators, riggers and company officials are aware of the actions required in the event of an accident as specified in the contract. 5. Signal persons used in construction work are qualified in accordance with 29 CFR 1926.1428. 6. Riggers are qualified in accordance with NAVFAC P-307, paragraph 11.1.k. 7. All personnel working on the job site have been trained to not stand under a load or in the fall zone of a suspended load unless specifically allowed by USACE EM 385-1-1.	
COMPANY OFFICIAL SIGNATURE	DATE
COMPANY OFFICIAL NAME/TITLE	
POST ON CRANE (OR ALTERNATE MACHINE) (IN CAB OR VEHICLE) (or in the contractor's on-site office for rigging operations)	

Figure P-1

CONTRACTOR CRANE OR RIGGING OPERATION CHECKLIST				
Before The Lift		YES	NO	N/A
1	Is the Certificate of Compliance, P-1, in the operator's cab (or in the contractor's on-site office for rigging operations) with the current operator's name listed?			
2	Is the crane/machine transited to and from the job site correctly? Are the OEM instructions for travel being followed?			
3	If this is a critical lift, is a lift plan provided and understood?			
3	If overhead power lines are in the vicinity, is a critical lift plan provided addressing the requirements of 29 CFR 1926.1407-1411?			
5	Does the operator know the weight of the load to be lifted (including rigging)?			
6	Is the load to be lifted within the crane/machine manufacturer's rated capacity in its present configuration?			
7	Are outriggers/stabilizers required and, if so, are they properly deployed and down?			
8	If outrigger/stabilizers are used, and the wheels are not off the ground is this the correct setup in accordance with the OEM?			
9	Is the crane/machine level and on firm ground, or if the ground is not firm are adequate supporting materials provided?			
10	If supporting materials are provided, is the entire surface of the outrigger/stabilizer pad supported and is the supporting material of sufficient strength to safely support the loaded outrigger/stabilizer pad?			
11	If the activity/location has specific ground loading restrictions or rules on ground bearing pressures, are those rules being followed?			
12	If outriggers/stabilizers are not used, is the crane/machine rated for on-rubber lifts by the OEM's load chart?			
13	Is the swing radius of the crane counterweight clear of people and obstructions and are accessible areas within the swing area barricaded to prevent injury or damage?			
14	Is the lift and rotation path clear of obstructions?			
15	Does rigging gear meet applicable ASME or host nation standards (e.g., ASME B30.9 for slings, B30.10 for hooks, B30.26 for rigging hardware such as shackles, swivel hoist rings, and eyebolts, B30.20 for below-the-hook lifting devices)?			
16	Is sling protection used to protect slings (especially synthetic slings) and equipment from damage due to abrasion and sharp corners and edges?			
17	Is the rigging gear used in accordance with its working load limit? Is the working load limit marked on the rigging gear?			
18	Was the rigging gear inspected prior to use?			
19	Is rigging gear undamaged and acceptable for the application?			
20	Are positive latching devices (or "mousing") used on crane and rigging hooks?			
21	If a mobile crane is used on a barge, are all rules of 29 CFR 1926.1437(construction) being followed?			
22	If a mobile crane is used on a barge are the outriggers/stabilizers blocked or are the crawlers traveling in a defined space as allowed by 29 CFR 1926.1437 (construction)?			
23	For floating cranes, are rules of 29 CFR 1915 (ship repair) or 29 CFR 1926.1437 (construction) being followed?			
24	If a multi-purpose machine, forklift, or construction equipment is being used, is there proof from the OEM (or qualified PE) that the machine is approved for suspended load lifting and is there a load chart?			

Figure P-2 (1 of 2)

		YES	NO	N/A
25	If a personnel lift is being performed with a crane or base mounted hoist, are all requirements of NAVFAC P-307, paragraph 11.1.g(5) for a crane or 11.1.m for a base mounted hoist being followed?			
	During The Lift			
26	Has the hook been centered over the load in such a manner to minimize swing and avoid side loading?			
27	Is the load well secured and balanced in the sling or lifting device after it is lifted a few inches for verification?			
28	Are taglines or other restraints being used, as necessary, for proper load control?			
29	Are personnel prevented from standing or passing under a suspended load?			
30	Are proper signals being used?			
31	Are radios being used for blind lifts?			
32	Is the operator paying full attention to and responding properly to the signals?			
33	Does the operator remain at the controls while the load is suspended?			
34	Is side loading occurring?			
35	Are personnel prevented from riding on a load?			
36	Are start and stop motions in a smooth fluid motion (no sudden acceleration or deceleration)?			
37	If pick and carry operations are allowed and performed, are OEM directions followed (e.g., rotation lock engaged, boom centered over front or rear)?			
38	When the crane/machine is left unattended, is it in a safe condition?			

Contractor:		Subcontractor:	
Location:		Date:	
Optional Information:			
Crane Operator:			
Make, Model, Serial Number of Crane:			
Load Weight:			
Crane Capacity at Pick Configuration:			
Rigging Equipment Capacity at Pick Configuration:			
Contract Number:			
Notes:			
Signature of Government Representative who completed this form:			

Figure P-2 (2 of 2)

CRANE AND RIGGING ACCIDENT REPORT					
Accident Category: ☐ Crane Accident ☐ Rigging Accident				☐ *Significant Accident	
Name and UIC of Reporting Activity:				Copy To: Navy Crane Center Bldg. 491 NNSY Portsmouth, VA 23709 Fax: 757-967-3808	
Name and UIC of Activity Responsible for the Accident:			Activity Name and UIC of Accident Location:		Report No:
Specific Location:			Accident Date:		Time:
Contractor Operation: ☐ Yes ☐ No If Yes, Contract No:			BOS Contractor ☐ Yes ☐ No		BOS Contractor Equip. ☐ Yes ☐ No
Crane No:	Crane Type:	Category:	Crane OEM:		
Crane Capacity:		Hoist Capacity:		Weight of Load on hook:	Weather:
Critical Lift/Critical Non-Crane Rigging Operation? ☐ Yes ☐ No			Ordinance Lift? ☐ Yes ☐ No		
Lost Work Days? ☐ Yes ☐ No		Fatality or Permanent Disability? ☐ Yes ☐ No		Material/ Property Cost Estimate:	
Accident Type (check all that apply): ☐ Personal Injury* ☐ Overload* ☐ Two Blocked* ☐ Power Line Contact* ☐ Dropped Load* ☐ Derail* ☐ Overturned Crane* ☐ Crane Collision ☐ Damaged Crane ☐ Damaged Load ☐ Load Collision ☐ Damaged Rigging Gear ☐ Lower Threshold Crane Accident ☐ Other: Specify _____ * signifies significant accident					
Cause of Accident (check all that apply): ☐ Improper Operation ☐ Equipment Failure ☐ Inadequate Visibility ☐ Improper Rigging ☐ Switch Alignment ☐ Inadequate Communication ☐ Track Condition ☐ Procedural Failure ☐ Other: Specify _____					
Responsibility (check all that apply): ☐ Crane Walker ☐ Rigger ☐ Rigger-in-Charge ☐ Operator ☐ Signal Person ☐ Maintenance ☐ Management/Supervision ☐ Other: Specify _____					
Crane Function: ☐ Travel ☐ Hoist ☐ Rotate ☐ Luffing ☐ Telescoping ☐ Other ☐ N/A					
Is this accident indicative of a recurring problem? ☐ Yes ☐ No If yes, list Accident Report Nos.: _____					
ATTACH COMPLETE AND CONCISE SITUATION DESCRIPTION AND CORRECTIVE/PREVENTIVE ACTIONS TAKEN AS ENCLOSURE (1). Include root cause and contributing factors. Assess damages and define responsibility. For equipment malfunction or failure, include specific description of the component and the resulting effect or problem caused by the malfunction or failure. List immediate and long-term corrective/preventive actions assigned and respective codes.					
INCLUDE: Printed Name, Code and Date.					
Preparer:		Phone:		E-mail:	
Code:		Date:			
Concurrence		Code:		Date:	
WHE Program Manager or Contracting Officer/designee (if Applicable)		Code:		Date:	
Certifying Official (Crane Accident Only):		Code:		Date:	
Contracting Officer's Representative (if Applicable)		Code:		Date:	

CRANE AND RIGGING ACCIDENT REPORT INSTRUCTIONS

1. Accident Category: Indicate either crane accident or rigging accident. Indicate if significant (see paragraph 12.3).
2. Reporting Activity/UIC: The activity and unit identification code responsible for reporting the accident in accordance with paragraph 12.6.2.
3. Activity Responsible for the Accident/UIC: Same as #2 above, or for NAVFACENGSSYSCOM, provide FEC level UIC.
4. Report No.: The activity assigned accident number (e.g., Activity UIC-FY-CA-01).
5. Accident Location UIC: The activity and unit identification code of where the event took place.
6. Specific Location: The detailed location where the event took place (e.g., building 213, drydock 5).
7. Accident Date: The date the accident occurred.
8. Time: The time (24-hour clock) the accident occurred (e.g., 1300).
9. Contractor Operation: Check yes or no. If yes, enter contract number.
10. BOS Contractor: Check yes or no.
11. BOS Contractor equipment: Check yes or no.
12. Crane No.: The activity assigned local crane number (e.g., PC5), if applicable.
13. Crane Type: The type of crane involved in the accident (e.g., mobile, bridge), if applicable.
14. Category: Identify category of crane (i.e., 1, 2, or 3), if applicable.
15. Crane OEM: The original equipment manufacturer of the crane (e.g., Samsung, Grove, P&H), if applicable.
16. Crane Capacity: The certified capacity of the crane (e.g., 120,000 pounds), if applicable.
17. Hoist Capacity: The capacity of the hoist involved in the accident at the max radius of the operation, if applicable.
18. Weight of Load on Hook: The weight of the load on the hook, if applicable.
19. Weather: The weather conditions at time of accident (e.g., wind, rain, cold).
20. Critical lift: Was the crane or rigging gear being used in a critical lift? Check yes or no.
21. Ordnance Lift: Was the crane or rigging gear being used in a lift governed by NAVSEA OP-5? Check yes or no.
22. Lost Workdays? Check yes or no.
23. Fatality or Permanent Disability: Check yes or no.
24. Material/Property Cost Estimate: Estimate total cost of damage resulting from the accident. (See OPNAV M-5102.1).
25. Accident Type: Check all that apply.
26. Cause of Accident: Check all that apply.
27. Responsibility: Check all that apply.
28. Crane Function: Check all functions in operation at time of accident. Check N/A if a rigging gear accident.
29. Is this a recurring problem? Check yes or no. If yes, list Accident Report numbers.
30. Preparer: Printed name must be provided.
31. Concurrences: Printed name must be provided.
32. Certifying Official (Crane Accidents Only): Printed name must be provided.
33. WHE Program Manager/Contracting Officer: Printed name must be provided when.
34. Contracting Officers representative: Printed name must be provided for contractor crane or rigging accidents.

Enclosure (1)

Brief Description: No more than one paragraph summarizing the resultant incident.

Root Cause and Detailed Description: Provide the relevant background in a descriptive timeline of preconditions leading up to the event, as well as a detailed description of the event.

Corrective Actions: List all short-term and long-term corrective actions that are taken to prevent recurrence of the incident. Short-term corrective actions are those actions taken that will allow return to work in short time frame. Long-term actions are more 'programmatic' in nature and typically include process revision, changes in training, 'mistake proofing', etc.

Note: Forms should be marked in accordance with the activity's security and marking policies.

CRANE AND RIGGING NEAR MISS REPORT INSTRUCTIONS

1. Near Miss Category: Indicate crane or rigging near miss.
2. Reporting Activity/UIC: The activity and unit identification code responsible for reporting the near miss in accordance with the guidance of paragraph 12.6.2.
3. Activity Responsible for the Near Miss/UIC: Same as #2 above, or for NAVFACENGSYSCOMs, provide the FEC level UIC.
4. Report No.: The activity assigned near miss number (e.g., Activity UIC-FY-CA-01).
5. Location UIC: The activity and unit identification code where the event took place.
6. Specific Location: The detailed location where the near miss took place (e.g., building 213, drydock 5).
7. Near Miss Date: The date the near miss occurred.
8. Time: The time (24-hour clock) the near miss occurred (e.g., 1300).
9. Is the responsible party a BOS Contractor? Check yes or no.
10. Is the responsible party a contractor other than a BOS contractor? Check yes or no. If yes, enter contract number.
11. Crane No.: The activity assigned crane number (e.g., PC-5), if applicable.
12. Crane Type: The type of crane involved in the near miss (e.g., mobile, bridge), if applicable.
13. Category: Identify category of crane (i.e., 1, 2, or 3), if applicable.
14. Crane OEM: The original equipment manufacturer of the crane (e.g., Samsung, Grove, P&H), if applicable.
15. Crane Capacity: The certified capacity of the crane (e.g., 120,000 pounds), if applicable.
16. Hoist Capacity: The certified capacity of the hoist involved in the near miss at the max radius of the operation, if applicable.
17. Weight of Load on Hook: If applicable, the weight of the load on the hook.
18. Weather: The weather conditions at time of accident (e.g., wind, rain, cold).
19. Critical lift: Was the crane or rigging gear being used in a critical lift?
20. Is this a recurring problem? Check yes or no. Identify any other similar near misses or accidents.
21. Brief Description: No more than one paragraph summarizing the resultant incident.
22. Root Cause and Detailed Description: Provide the relevant background in a descriptive timeline of preconditions leading up to the event, as well as a detailed description of the event.
23. Corrective Actions: List all short-term and long-term corrective actions that were/will be taken to prevent recurrence of the incident. Short-term corrective actions are those actions taken that will allow return to work in short time frame. Long-term actions are more 'programmatic' in nature and typically include process revision, changes in training, 'mistake proofing', etc.

Note: Forms should be marked in accordance with the activity's security and marking policies.

Figure 12-2 (2 of 2)

CRANE AND RIGGING UNPLANNED OCCURRENCE REPORT					
Unplanned Occurrence Category: ☐ Crane ☐ Rigging				☐ *Injury/Substantial	
Reporting Activity:				Copy To: Navy Crane Center Bldg. 491 NNSY Portsmouth, VA 23709 Fax: 757-967-3808	
UIC:					
Activity Responsible:			Report No:		
UIC:			Event Location UIC:		Event Date: Time:
Specific Location:			BOS Contractor ☐ Yes ☐ No		BOS Contractor Equip. ☐ Yes ☐ No
Crane No:	Crane Type:	Category:	Crane OEM:		
Crane Capacity:		Hoist Capacity:		Weight of Load on hook:	Weather:
Lost Work Days? ☐ Yes ☐ No		Fatality or Permanent Disability? ☐ Yes ☐ No		Material/ Property Cost Estimate:	
Unplanned Occurrence Type (check all that apply):					
☐ Personal Injury* ☐ Substantial Damage* ☐ Crane Collision ☐ Damaged Crane ☐ Damaged Rigging Gear ☐ Equipment Failure ☐ Other: Specify _____					
Cause of Unplanned Occurrence (check all that apply):					
☐ Foreign Material Contamination ☐ Inadequate Visibility ☐ Improper Rigging Inspection ☐ Hazardous Energy Control ☐ Inadequate Communication ☐ Poor Mechanic Technique ☐ Unclear Procedure ☐ Procedure Violation ☐ Other: Specify _____					
Responsibility (check all that apply):					
☐ Mechanic ☐ Electrician ☐ Operator ☐ Rigger ☐ Engineer ☐ Inspector ☐ Management/Supervision ☐ Other: Specify _____					
Is this event indicative of a recurring problem? ☐ Yes ☐ No					
If yes, list Report Nos.: _____					
ATTACH COMPLETE AND CONCISE SITUATION DESCRIPTION AND CORRECTIVE/PREVENTIVE ACTIONS TAKEN AS ENCLOSURE (1). Include root cause and contributing factors. Assess damages and define responsibility. For equipment malfunction or failure, include specific description of the component and the resulting effect or problem caused by the malfunction or failure. List immediate and long-term corrective/preventive actions assigned and respective codes.					
INCLUDE: Printed Name, Code and Date.					
Preparer:		Phone:		E-mail:	
				Code:	Date:
Concurrence				Date:	
WHE Program Manager or Contracting Officer/designee (if Applicable)				Date:	
Certifying Official (Crane Event Only):					
Contracting Officer's Representative (if Applicable)					

Figure 12-3 (1 of 2)

CRANE AND RIGGING UNPLANNED OCCURRENCE REPORT INSTRUCTIONS

1. Unplanned Occurrence Category: Indicate either crane or rigging. Indicate if injury or substantial damage occurred.
2. Reporting Activity/UIC: The activity and unit identification code responsible for reporting the event in accordance with the guidance of paragraph 12.6.2.
3. Activity Responsible for the Event UIC: Same as #2 above, or for NAVFACENGSYSCOMs, provide the FEC level UIC.
4. Report No.: The activity assigned event number (e.g., Activity UIC-FY-CA-01).
5. Event Location UIC: The activity and unit identification code of where the event took place.
6. Event Date: The date the event occurred.
7. Time: The time (24-hour clock) the event occurred (e.g., 1300).
8. Specific Location: The detailed location where the event took place (e.g., building 213, drydock 5).
9. BOS Contractor: Check yes or no.
10. BOS Contractor equipment: Check yes or no.
11. Crane No.: The activity assigned local crane number (e.g., PC5), if applicable.
12. Crane Type: The type of crane involved in the event (e.g., mobile, bridge), if applicable.
13. Category: Identify category of crane (i.e., 1, 2, or 3), if applicable.
14. Crane OEM: The original equipment manufacturer of the crane (e.g., Samsung, Grove, P&H), if applicable.
15. Crane Capacity: The certified capacity of the crane (e.g., 120,000 pounds), if applicable.
16. Hoist Capacity: The capacity of the hoist involved in the event at the max radius of the operation, if applicable.
17. Weight of Load on Hook: The weight of the load on the hook, if applicable.
18. Weather: The weather conditions at time of the event (e.g., wind, rain, cold).
19. Lost Work Days? Check yes or no.
20. Fatality or Permanent Disability: Check yes or no.
21. Material/Property Cost Estimate: Estimate total cost of damage resulting from the event.
22. Unplanned Occurrence Type: Check all that apply.
23. Cause of Unplanned Occurrence: Check all that apply.
24. Responsibility: Check all that apply.
25. Is this a recurring problem? Check yes or no. If yes, list report numbers.
26. Preparer: Printed name must be provided.
27. Concurrences: Printed name must be provided.
28. Certifying Official: Printed name must be provided.

Enclosure (1)

Brief Description: No more than one paragraph summarizing the resultant incident.

Root Cause and Detailed Description: Provide the relevant background in a descriptive timeline of preconditions leading up to the event, as well as a detailed description of the event.

Corrective Actions: List all short-term and long-term corrective actions that are taken to prevent recurrence of the incident. Short-term corrective actions are those actions taken that will allow return to work in short time frame. Long-term actions are more 'programmatic' in nature and typically include process revision, changes in training, 'mistake proofing', etc.

Note: Forms should be marked in accordance with the activity's security and marking policies.

Figure 12-3 (2 of 2)

NAVFAC MIDLANT CRITICAL LIFT PLAN																																																																				
Date:		Qualified Person:																																																																		
Location:																																																																				
<i>Contractor is required to fill sections A.thru F. prior to submitting plan. Sections G.- H to be completed by PWD Maine ET</i>																																																																				
A. TOTAL LOAD 1. Load Weight lbs 2. Wt. of Aux. Block lbs 3. Wt. of Main Block lbs 4. Wt. of Lifting Beam lbs 5. Wt. of Sling/Shackles lbs 6. Wt. of Jib/Ext. (erected/stowed) lbs 7. Wt. of Hoist Rope lbs 8. Other: lbs TOTAL WEIGHT		E. CRANE PLACEMENT (Mobile Cranes Only) 1. Maximum Bearing Pressure PSF Note: Bearing Pressure Calculations must be attached on Page 3. 2. Ground Conditions Suitable for Load? YES / NO Note: Ground Condition Calculations must be attached on Page 3. 3. High Voltage or Electrical Hazards? YES / NO Note: If Electrical Hazards are present they must be shown on Page 4. 4. Obstructions to Lift or Swing? YES / NO Note: If Obstructions are present they must be shown on Page 4. 5. Travel with Load Required? YES / NO 6. Other?																																																																		
B. CRANE 1. Type of Crane <u>Mobile Hydraulic Truck</u> 2. Maximum Crane Capacity lbs. 3. Radius (Maximum) ft. 4. Radius (Minimum) ft. 5. Boom Length (Maximum) ft. 6. Boom Length (Minimum) ft. 7. Crane Capacity (Max Radius) lbs. 8. Crane Capacity (Min Radius) lbs. 9. Boom Angle (Maximum) deg. 10. Boom Angle (Minimum) deg. 11. Gross Load of Crane lbs. 12. Lift is _____ % of the Crane's rated capacity 13. If Jib/Ext. is to be used: Length ft. Offset ft. 14. Rated Capacity of Jib/Ext. lbs		F. OPERATOR QUALIFICATIONS 1. Certified Operator? YES / NO 2. Option? YES / NO 3. Certified for Type, Class & Capacity? YES / NO 4. Designated in writing by emp _____ <table border="1" style="width: 100%; border-collapse: collapse; margin-top: 5px;"> <thead> <tr> <th style="text-align: left; padding: 2px;">G. COMPLETED IN FIELD:</th> <th style="text-align: center; padding: 2px;">(YES)</th> <th style="text-align: center; padding: 2px;">NO</th> <th style="text-align: center; padding: 2px;">PWD ET</th> </tr> </thead> <tbody> <tr><td>1. Crane Inspected</td><td></td><td></td><td></td></tr> <tr><td>2. Rigging Inspected</td><td></td><td></td><td></td></tr> <tr><td>3. Crane Set-up</td><td></td><td></td><td></td></tr> <tr><td>4. Overhead Hazard Check</td><td></td><td></td><td></td></tr> <tr><td>5. Swing Check</td><td></td><td></td><td></td></tr> <tr><td>6. Counterweight Check</td><td></td><td></td><td></td></tr> <tr><td>7. Operator Qualifications</td><td></td><td></td><td></td></tr> <tr><td>8. Signal Person Qualifications</td><td></td><td></td><td></td></tr> <tr><td>9. Rigger Qualifications</td><td></td><td></td><td></td></tr> <tr><td>10. Load Chart in Crane</td><td></td><td></td><td></td></tr> <tr><td>11. Load Test</td><td></td><td></td><td></td></tr> <tr><td>12. Tag Lines</td><td></td><td></td><td></td></tr> <tr><td>13. Wind Conditions</td><td></td><td></td><td></td></tr> <tr><td>14. Traffic Hazard Check</td><td></td><td></td><td></td></tr> <tr><td>15. Site Control</td><td></td><td></td><td></td></tr> </tbody> </table>			G. COMPLETED IN FIELD:	(YES)	NO	PWD ET	1. Crane Inspected				2. Rigging Inspected				3. Crane Set-up				4. Overhead Hazard Check				5. Swing Check				6. Counterweight Check				7. Operator Qualifications				8. Signal Person Qualifications				9. Rigger Qualifications				10. Load Chart in Crane				11. Load Test				12. Tag Lines				13. Wind Conditions				14. Traffic Hazard Check				15. Site Control			
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C. HOIST ROPE 1. # of Parts 2. Rope Diameter 3. Capacity	Main	Aux 1	Aux 2	D. RIGGING 1. Hitch Type(s) 2. No. of Slings: Size: _____ 3. Sling Type: 4. Sling Assembly Capacity: lbs. 5. Shackle Size(s): 6. Shackle Rated Capacity(s) lbs.																																																																
H. SIGNATURES IN FIELD: 1. Crane Operator 2. Rigger 3. Signal Person 4. Lift Supervisor 5. PWD ET 6. Other																																																																				

EM 385-1-1
XX Jul 14

U.S. Army Corps of Engineers

CRITICAL LIFT PLAN

For use of this form, see EM 385-1-1, Section 16. Proponent agency is Crane HHWG.

SITE PLAN

Show here or attach site plan and sequencing

